

National University of Computer & Emerging Sciences

MT2005 – Probability and Statistics

Homework # 06

Gamma Distribution

1. The probability density function of gamma distribution is:

$$f(x) = \frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}, \quad x \geq 0,$$

where

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx, \quad \alpha > 0$$

is the gamma function for integer α .

- a) By considering the duality between the poisson process (counting events in a fixed time) and the gamma distribution (waiting time for α events), derive the probability density function of gamma distribution. (*Hint*: Relate the cumulative distribution function $P(X \leq x)$ to the Poisson probability of observing at least α events by time x , then differentiate.)
 - b) Derive the $E[X]$, $E[X^2]$ and $\text{Var}(X)$ of the gamma distribution without using the MGF. (*Hint*: First make the substitution $u = \lambda x$ to simplify the integral. Then, use the property $\Gamma(\alpha + 1) = \alpha \Gamma(\alpha)$.)
 - c) Find the mode of gamma distribution.
2. The number of automobiles that arrive at a certain intersection per minute has a Poisson distribution with a mean of 5. Interest centers around the time that elapses before 10 automobiles appear at the intersection. What is the probability that more than 2 minutes elapse before 10 cars arrive?
3. For an electrical component with a failure rate of once every 5 hours, it is important to consider the time that it takes for 2 components to fail.
- a) Assuming that the gamma distribution applies, what is the mean time that it takes for 2 components to fail?
 - b) What is the probability that 12 hours will elapse before 2 components fail?